

Homotopy.

Def. Let $f, g: X \rightarrow Y$ be continuous maps.

We said that f and g are Homotopic if there exists a Continuous map

$$F: X \times I \rightarrow Y \text{ such that } \begin{cases} F(x, 0) = f(x), & \forall x \in X. \\ F(x, 1) = g(x), & \forall x \in X. \end{cases}$$

This F is called a Homotopy, and in such a case, denote $f \simeq g$.

Thm. Let $C(X, Y) \stackrel{\text{def}}{=} \{f: X \rightarrow Y \mid f \text{ is continuous}\}$,

Define a relation on $C(X, Y)$ as: $f \sim g \Leftrightarrow f \simeq g$.

Then, the Homotopy is equivalence relation.

Proof. Reflexivity: Given $f \in C(X, Y)$, define

$$F: X \times I \rightarrow Y: (x, t) \mapsto f(x)$$

Then, since $F^{-1}[V] = f^{-1}[V] \times I$, F is continuous such that $F(x, 0) = F(x, 1) = f(x)$.

Symmetric. Let $f \simeq g$ via $F: X \times I \rightarrow Y$. Define $G: X \times I \rightarrow Y: (x, t) \mapsto F(x, 1-t)$

Then, $G(x, 0) = F(x, 1) = g(x)$ and $G(x, 1) = F(x, 0) = f(x)$.

Transitivity. Let $f \simeq g$ via F and $g \simeq h$ via G . Define

$$H: X \times I \rightarrow Y: (x, t) \mapsto \begin{cases} F(x, 2t) & \text{if } 0 \leq t \leq \frac{1}{2} \\ G(x, 2t-1) & \text{if } \frac{1}{2} \leq t \leq 1. \end{cases}$$

Then, since $[0, \frac{1}{2}]$ and $[\frac{1}{2}, 1]$ are closed with $F(x, 1) = g(x) = G(x, 0)$, therefore, the Pasting lemma gives that the H is Homotopy from f to h . So $f \simeq h$.

Def. Let X and Y be topological space.

We said that X and Y are Homotopy equivalent if there exists two Continuous map $f: X \rightarrow Y$ and $g: Y \rightarrow X$ such that $g \circ f \simeq \text{id}_X$ and $f \circ g \simeq \text{id}_Y$.

In such a case, denote $X \simeq Y$.

And, such a $f: X \rightarrow Y$ is called a Homotopy equivalence, and such a $g: Y \rightarrow X$ is called a Homotopy inverse.

Note: Trivially, Homeomorphism is Homotopy equivalence. Therefore,

$X \cong Y \Rightarrow X \simeq Y$, but the Converse does not hold.

Counterexample: Consider two subspaces $S', \mathbb{R}^2 \setminus \{0\} \subseteq \mathbb{R}^2$.

First, since S' is closed and bounded, so it is Compact.

And, since $\{0\}$ is closed, $\mathbb{R}^2 \setminus \{0\}$ is open, so it is not Compact. So, $S' \not\simeq \mathbb{R}^2 \setminus \{0\}$.

Next, to show that $S' \simeq \mathbb{R}^2 \setminus \{0\}$, Consider that an inclusion

$$f: S' \hookrightarrow \mathbb{R}^2 \setminus \{0\}; x \mapsto x \quad \text{and}$$

$$g: \mathbb{R}^2 \setminus \{0\} \rightarrow S'; x \mapsto \frac{x}{\|x\|}.$$

Then, both f and g are Continuous, and $g \circ f = \text{id}_{S'}$.

And, $(f \circ g): \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}^2 \setminus \{0\}; x \mapsto x/\|x\|$ is Homotopic to $\text{id}_{\mathbb{R}^2 \setminus \{0\}}$ via

$$F: [\mathbb{R}^2 \setminus \{0\}] \times [0, 1] \rightarrow \mathbb{R}^2 \setminus \{0\}; (x, t) \mapsto (1-t) \cdot \frac{x}{\|x\|} + t \cdot x.$$

which satisfies $F(x, 0) = x/\|x\|$ and $F(x, 1) = x$, so Prove.

Contractible

Def.

A space that homotopy equivalent to a point is called Contractible.

Thm. Let X be a Topological space. Then, TFAE:

a). X is Contractible. (Homotopy equivalent to a point)

b). $\text{id}_X: X \rightarrow X: x \mapsto x$ is null-homotopic.

c). Given $x \in X$, $C_x \simeq \text{id}_X$ where $C_x: X \rightarrow X: z \mapsto x$.

Proof.

Assume a). That is, there exist two maps

$f: X \rightarrow \{p\}$ and $g: \{p\} \rightarrow X$ such that $f \circ g \simeq \text{id}_p$ and $g \circ f \simeq \text{id}_X$.

Then, since $f: X \rightarrow \{p\}$ must be a Constant Mapping $x \mapsto p$,

$g \circ f(x) = g(p)$ for all $x \in X$, so $g \circ f$ is Constant, so id_X is null-homotopic.

It proved b).

Assume b). Let $C: X \rightarrow X: z \mapsto x$ be a Constant map such that $C \simeq \text{id}_X$, via

$H: X \times I \rightarrow X$ such that $H(z, 0) = z$ and $H(z, 1) = x$.

Let $x \in X$ be fixed. Define a path $p: [0, 1] \rightarrow X: t \mapsto H(x, t)$.

Then, $p(0) = H(x, 0) = x$ and $p(1) = H(x, 1) = x$.

Now, $F: X \times I \rightarrow X: (z, t) \mapsto p(t)$ is the Homotopy to $C \simeq C_x$. Now, by transitivity, $C \simeq \text{id}_X$ and $C \simeq C_x$ implies that $C_x \simeq \text{id}_X$. It proved c).

Assume c). Let $f: X \rightarrow \{p\}: x \mapsto p$ be a map and let $x \in X$ be given

Then, a map $g: \{p\} \rightarrow X: p \mapsto x$ satisfies

$$f \circ g = \text{id}_p \text{ and } g \circ f = C_x.$$

By c), $C_x \simeq \text{id}_X$, so $g \circ f \simeq \text{id}_X$. So proved $\square$

Corollary. Contractible space is Path-Connected.

Proof. Suppose that X is Contractible space. Let $x, y \in X$ be given,

Then, $C_x \simeq \text{id}_X \simeq C_y$, so there exists a Homotopy

$$F: X \times I \rightarrow X \text{ such that } \begin{cases} F(z, 0) = C_x(z) = x \text{ for all } z \in X \\ F(z, 1) = C_y(z) = y \text{ for all } z \in X. \end{cases}$$

Now, $p: I \rightarrow X: t \mapsto F(x, t)$ is a path from x to y . $\square$

Basic Propositions.

Prop. Let $f, g: X \rightarrow Y$ be continuous and $f \simeq g$.

If f is Homotopy equivalence, then g is also Homotopy equivalence

Proof. Let $h: Y \rightarrow X$ be the Homotopy inverse of f .

$$h \circ f \simeq \text{id}_X \text{ and } f \circ h \simeq \text{id}_Y.$$

Meanwhile, let $F: X \times I \rightarrow Y$ be the Homotopy from f to g .

Then, $H: X \times I \rightarrow X: (x, t) \mapsto h(F(x, t))$ satisfying $\begin{cases} H(x, 0) = h(f(x)) \\ H(x, 1) = h(g(x)) \end{cases}$

So, $h \circ f \simeq h \circ g$, thus $h \circ g \simeq \text{id}_X$. And, define

$$G: Y \times I \rightarrow Y: (y, t) \mapsto F(h(y), t) \text{ satisfying } \begin{cases} G(y, 0) = f(h(y)) \\ G(y, 1) = g(h(y)) \end{cases}$$

So, $f \circ h \simeq g \circ h$, thus $g \circ h \simeq \text{id}_Y$. So proved. $\square$

Prop. Let $f, g: X \rightarrow \prod_{\alpha \in \Lambda} Y_\alpha$ be continuous maps. Then, for projection π_α ,

$f \simeq g$ if and only if for all $\alpha \in \Lambda$, $\pi_\alpha \circ f \simeq \pi_\alpha \circ g$.

Proof. Suppose that there is a Homotopy $F: X \times I \rightarrow \prod_{\alpha \in \Lambda} Y_\alpha$. (That is, $f \simeq g$).

Then, the projection $\pi_\beta: \prod_{\alpha \in \Lambda} Y_\alpha$ induces a Homotopy

$$(\pi_\beta \circ F): X \times I \rightarrow Y_\beta: (x, t) \mapsto \pi_\beta(F(x, t)).$$

Satisfying $(\pi_\beta \circ F)(x, 0) = (\pi_\beta \circ f)(x)$ and $(\pi_\beta \circ F)(x, 1) = (\pi_\beta \circ g)(x)$, so that $\pi_\beta \circ f \simeq \pi_\beta \circ g$. Conversely, suppose that for all $\alpha \in \Lambda$, $\pi_\alpha \circ f \simeq \pi_\alpha \circ g$.

For each $\alpha \in \Lambda$, let $F_\alpha: X \times I \rightarrow Y_\alpha$ be the Homotopy from $\pi_\alpha \circ f$ to $\pi_\alpha \circ g$.

Define $F: X \times I \rightarrow \prod_{\alpha \in \Lambda} Y_\alpha: (x, t) \mapsto (F_\alpha(x, t))_{\alpha \in \Lambda}$

Then, by the property of product space, F is continuous, so, it is a Homotopy from f to g .

Prop. If $X_\alpha \simeq Y_\alpha$ for all $\alpha \in \Lambda$, then $\prod_{\alpha \in \Lambda} X_\alpha \simeq \prod_{\alpha \in \Lambda} Y_\alpha$.

$\square$

Prop. If Y is Contractible, then $X \times Y \cong X$.

Proof. Let $P = \{p\}$ be a one-point set space which is Homotopy equivalent to Y . That is, there exist two Continuous maps $f: P \rightarrow Y$ and $g: Y \rightarrow P$ satisfy

$$f \circ g \cong \text{id}_Y \quad \text{and} \quad g \circ f \cong \text{id}_P.$$

Actually, $g \circ f$ must equals to id_P . To show $X \times Y \cong X \times P$, define the Homotopy equivalence

$$(\text{id}_X \times g): X \times Y \rightarrow X \times P; (x, y) \mapsto (x, p)$$

with the Homotopy inverse

$$(\text{id}_X \times f): X \times P \rightarrow X \times Y; (x, p) \mapsto (x, f(p)).$$

Then, $(\text{id}_X \times g) \circ (\text{id}_X \times f) \equiv \text{id}_X \times \text{id}_P = \text{id}_{X \times P}$.

Meanwhile, to show $(\text{id}_X \times f) \circ (\text{id}_X \times g) \cong \text{id}_{X \times Y}$, let

$$F: Y \times I \rightarrow Y \text{ such that } F(y, 0) = (f \circ g)(y) \text{ and } F(y, 1) \equiv \text{id}_Y.$$

be the Homotopy. Now, the map

$$H: (X \times Y) \times I \rightarrow X \times Y; ((x, y), t) \mapsto (x, F(y, t))$$

Finally, $H(\cdot, 0) = (x, (f \circ g)(y))$ and $H(\cdot, 1) = (x, \text{id}_Y(y)) = (x, y)$,

so $(\text{id}_X \times f) \circ (\text{id}_X \times g) \cong \text{id}_{X \times Y}$. Consequently, $X \times Y \cong X \times P$ and since $X \cong X \times P$ thus $X \times Y \cong X \times P \cong X$. $\square$

Prop. Let X be any space, and Y be a contractible.
Then, for any continuous map $f, g: X \rightarrow Y$, $f \simeq g$.
Moreover, every continuous map $h: X \rightarrow Y$ is null-homotopic.

Prop. Let X be a Topological space, and let

$$CX = X \times I / X \times \{1\}$$

be a Cone, where $I = [0, 1]$.

Then, CX is Contractible. Moreover, for any space Y and continuous $f: X \rightarrow Y$, $f: X \rightarrow Y$ is null-homotopic if and only if there is an extension $\tilde{f}: CX \rightarrow Y$.

Proof. Let $P = \{P_0\}$ for fixed $P_0 = [(x_0, 1)] \in CX$.

$f: CX \rightarrow P: x \mapsto P_0$ is a Constant map, so continuous with homotopy inverse $g: P \rightarrow CX: P_0 \mapsto P_0$ so that $f \circ g = \text{id}_P \simeq \text{id}_P$.

And, $g \circ f(x) = P_0$ is a Constant map, let's show $g \circ f \simeq \text{id}_{CX}$.

Construct a Homotopy between $g \circ f$ and id_{CX} . Define

$$H: CX \times I \rightarrow CX; [(x, s), t] \mapsto [(x, (1-t) \cdot s + t)]$$

Then, $H([(x, s)], 0) = [(x, s)]$ and $H([(x, s)], 1) = [(x, 1)]$

(Specifically, H is well-defined Continuous map,

$$(Note: H([(x, s)], 0) = [(x, s)])$$

